Yile (Wayne) Wang

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SUMMARY

Computational neuroscientist who designs and ships AI agent systems. Creator and maintainer of LLM-for-Zotero (**3,100+ GitHub stars, 650K+ plugin downloads, 100+ releases**), a TypeScript research agent with an adaptive agent runtime, multi-provider model layer, context budgeting, and verified, undoable actions. Research on brain-aligned deep networks (*Communications Biology*) and on how representational drift shapes neural decoding (first-author ICLR 2027 submission).

EDUCATION

The University of Texas at Dallas — **Richardson, TX**
Ph.D., Cognition and Neuroscience (GPA: 3.94/4) — Aug 2021 – Aug 2027 (expected)
M.S., Applied Cognition and Neuroscience, Computational Modeling track (GPA: 3.93/4) — Aug 2019 – May 2021

Southern Medical University — **Guangzhou, China**
B.S., Psychology — Aug 2014 – May 2018

RESEARCH & WORK EXPERIENCE

Doctoral Researcher, The University of Texas at Dallas — Aug 2021 – Present

Representational drift: biologically constrained neural network modeling (with Kaushik Lakshminarasimhan)

- Designed and trained neural networks to explain representational drift, translating biological constraints into model architecture, learning rules, and training strategies.
- Adapted learning rules, model parameters, and training procedures to capture biological mechanisms while meeting modeling objectives.
- Developed an efficient-coding account of representational drift; first-author paper under review at **ICLR 2027**.
- Built a biologically plausible **downstream decoding model** that uses **online statistical learning** to compensate for representational drift. Validated the model on the longitudinal THINGS-EEG2 dataset; manuscript **under review**.

Brain-aligned vision models, ReAlnet (with Zitong Lu and Julie Golomb, Ohio State University)

- Co-designed a multi-layer image-to-brain encoding framework that jointly trains ImageNet-pretrained CORnet-S for object classification and EEG generation. Combined mean-squared-error (MSE) and contrastive losses for EEG generation, yielding **10 individualized models** aligned to each subject's EEG.
- ReAlnets outperformed the **baseline model, CORnet-S**, in EEG representational similarity across all four visual layers (**up to 40% relative improvement at peak**). Gains extended to unseen fMRI data and human behavioral benchmarks on Brain-Score.
- Built the **PyTorch training and validation pipeline**, including Brain-Score behavioral evaluation and representational similarity analysis. Ran control experiments (**loss ablations, unpaired and scrambled EEG**) isolating the contribution of neural alignment.
- Extended the framework to **ResNet-18**, achieving significant improvements in EEG representational similarity and demonstrating applicability beyond the original architecture.

Publications: *Communications Biology* (2026; earlier version at an ICLR 2024 workshop) and *Cognitive Neurodynamics* (2025).

Computational modeling and biomarker discovery in Alzheimer's disease

- Predicted conversion from mild cognitive impairment (MCI) to Alzheimer's disease **4 years before clinical diagnosis with 80% specificity**. Used computational biomarkers from individualized brain-network models and multivariate discriminant analysis.
- Built **individualized digital brain twins in The Virtual Brain**, using dynamical systems constrained by structural MRI and diffusion imaging. Applied mean-field modeling and criticality analysis to characterize collective brain dynamics in MCI and Alzheimer's disease.
- Used **Python/pandas** to clean and integrate **high dimensional, multimodal data from hundreds of patients**, spanning MRI, cognitive assessments, and behavioral measures. Applied **multi-variant clustering analysis and dimensionality reduction** to develop candidate biomarkers for disease classification and progression prediction.

HPC Research Assistant (NSF-funded), Office of Information Technology, UT Dallas — Aug 2023 – May 2024

- Deployed a brain simulation platform on a shared server with a **PostgreSQL backend** for multi-user simulation and data management.
- Built a **CUDA-enabled container** and integrated GPU-accelerated fiber-orientation estimation with **Slurm job scripts** into a validated UK Biobank MRI pipeline. Achieved **approximately 100× speedup** for fiber-orientation estimation compared with the CPU implementation.

OPEN-SOURCE SOFTWARE

LLM-for-Zotero: Open-source AI research agent (GitHub) — *TypeScript, 2026 – present*

Creator, lead architect, and maintainer

- **Product ownership:** Created and shipped an AI research agent with **3,100+ GitHub stars and 650K+ plugin downloads**. Lead system architecture, implementation, **testing**, and ongoing maintenance.
- **Agent architecture:** Designed an **adaptive agent loop** that coordinates planning, tool execution, and continuation based on observed progress. Built an **agent harness** for instruction assembly, tool access, and runtime guardrails, including context budgets, input validation, and repeated-failure detection.
- **RAG and context engineering:** Combined **retrieval-augmented generation (RAG) and agentic search** for research across large document libraries. Optimized retrieval through library indexing, hybrid keyword and semantic ranking, and evidence caching. Designed automatic **context compaction** and source-coverage tracking to preserve reusable evidence across long research sessions.
- **Execution reliability:** Designed **state machines and action contracts** with explicit **component ownership**. The Zotero host application controls permissions, execution, and verification of saved results. Implemented **asynchronous job orchestration for batch processing**, with persisted checkpoints, cancellation, and recovery from interrupted runs. Supported undo for saved library changes.
- **Model and agent integration:** Integrated cloud and local models, the **Claude Agent SDK** through a companion bridge, and **Codex app-server**. Built a **Model Context Protocol (MCP) server** exposing Zotero tools to external agents. Optimized prompt caching through stable prompt prefixes and managed session reuse for continued conversations.

- **Testing and delivery:** Built **workflow test suites and behavioral evaluation suites** covering tool selection, permission handling, evidence quality, and verification of saved data inside Zotero. Designed automated testing, build, and publishing workflows with **GitHub Actions**, supporting **100+ releases**.

SKILLS

Programming & data: Python, pandas, TypeScript, R, MATLAB, Bash
AI agents & retrieval: Adaptive agent loops, agent harnesses, retrieval-augmented generation (RAG), agentic search, context compaction, prompt caching
Software architecture & integration: State machines, asynchronous workflows, action contracts, recovery mechanisms, Model Context Protocol (MCP), Claude Agent SDK, Codex app-server
Machine learning & statistics: PyTorch, convolutional neural networks, contrastive learning, dimensionality reduction, discriminant analysis, cross-validation
Computational modeling: Dynamical network simulation, biologically constrained neural networks, mean-field modeling, criticality analysis, information theory, Markov chain Monte Carlo (MCMC)
Testing & evaluation: Test-driven development (TDD), workflow test suites, regression testing
Infrastructure: SQLite, PostgreSQL, Docker, CUDA, Slurm, high-performance computing, Git, GitHub Actions, AWS, Google Cloud

SELECTED PUBLICATIONS

143 citations (Google Scholar); research areas: **machine learning, representational drift, and whole-brain modeling**.

Peer-reviewed journal articles

- **Wang, Y.**, Lu, Z., & Lakshminarasimhan, K. (2026). Coordinated drift of image representations in human EEG. [under review]
- Lu, Z., **Wang, Y.**, & Golomb, J. D. (2026). Achieving more human brain-like vision via human EEG representational alignment. *Communications Biology*, 9, 463. doi:10.1038/s42003-026-09685-w
- Lu, Z., & **Wang, Y.** (2025). Teaching CORnet human fMRI representations for enhanced model–brain alignment. *Cognitive Neurodynamics*, 19(1), 61.

Conference papers and presentations

- **Wang, Y.**, & Lakshminarasimhan, K. (2026). An efficient coding account of representational drift. *International Conference on Learning Representations (ICLR 2027)*. [under review]
- Hosseinpourkhoshkbari, R., Huang, W., **Wang, Y.**, Helabad, A., Chen, Z., & Golden, R. (2026). Human-likeness is multi-dimensional: Evaluating LLM-generated virtual patients against simulated patient dialogue. *Conference on Language Modeling (COLM 2026)*. [accepted]
- **Wang, Y.**, & Lakshminarasimhan, K. (2026). The geometry of drifting spatial representations. *Cognitive Computational Neuroscience (CCN 2026)*. [accepted poster]
- Lu, Z., **Wang, Y.**, & Golomb, J. D. (2024). ReAlnet: Achieving more human brain-like vision via human neural representational alignment. *ICLR 2024 Workshop*. [accepted]

HONORS & AWARDS

Marlane Miller Visionary New Scientist Award	Oct 2023
Society for Neuroscience Trainee Professional Development Award	Nov 2022
UT Dallas Master's Research Fellowship	Jul 2021